MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2013
Higher 2

CANDIDATE NAME	

CIVICS GROUP	1	2	S				

INDEX NUMBER		

H2 BIOLOGY **9648/03**

Applications Paper and Planning Question **23 September 2013**

Paper 3 **2 hours**

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer **all** questions.

At the end of the examination,
1. Fasten all your work securely together.
2. Hand in the following separately:
   - Structured questions 1 – 3
   - Planning question
   - Essay question

The number of marks is given in brackets [ ] at the end of each question or part question.

ANSWER SCHEME

For examiner's Use	
1	/ 15
2	/ 16
3	/ 9
4 Planning	/ 12
5 Essay	/ 20
Total	/ 72

This paper consists of __ printed pages.

[Turn over]

Answer **all** questions.

QUESTION 1

pCMV6-XL5 is a plasmid with a multiple cloning site (MCS) that lies downstream of the CMV promoter (Fig. 1.1). This plasmid can be inserted into both eukaryotic and prokaryotic host cells. The arrows denote the direction in which the genes are transcribed.

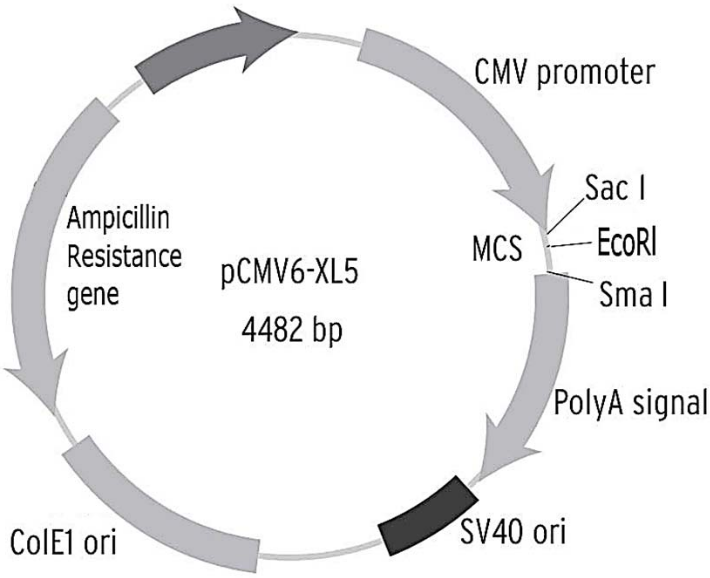


Fig. 1.1

(a) State the function of the CMV promoter in pCMV6-XL5. [1]

- It is where **RNA polymerase binds** to **initiate transcription** of the human growth hormone gene inserted into the multiple cloning site (downstream of the promoter).

An artificially-synthesised human growth hormone (hGH) gene with flanking *Hind*III restriction site sequences was created. The restriction sites for the restriction enzymes *Hind*III and *Sac*I are shown in Fig. 1.2.

HindIII

5′ . . . A▼A G C T T . . . 3′
3′ . . . T T C G A▲A . . . 5′

HindIII

SacI

5′ . . . G▼A G C T C . . . 3′
3′ . . . C T C G A▲G . . . 5′

SacI

▲ - point at which restriction enzymes cleave DNA.

Fig. 1.2

(b) Using information in Fig. 1.1 and 1.2, explain how the hGH gene can be inserted into pCMV6-XL5. [3]

- Cleave **gene** using ***Hind*III** to generate **sticky ends**.
- Cleave **plasmid** with ***Sac*I** to generate **complementary sticky ends** to the *Hind*III sticky ends flanking the hGH gene.
- Mix the cleaved gene and plasmid together and add **DNA ligase** to **seal the nicks** / **form phosphodiester bond** between gene and plasmid.

(c) State the conditions required for artificial transformation of *E. coli* cells with the plasmid vector. [2]

- Addition of **calcium chloride** solution to the mixture of E. coli and plasmid.
- Subject mixture to **heat shock** in a 42°C water bath for 2 minutes (or less).

(d) Describe the steps required to select bacterial colonies containing the human growth hormone gene after artificial transformation has taken place. [5]

1. Spread transformed bacteria onto agar/nutrient medium **plates containing ampicillin**.
2. **Bacterial colonies** on the ampicillin plate are **transferred** onto a **nitrocellulose membrane**.
3. Add **sodium hydroxide** to the filter to **lyse bacterial cells** and **denature dsDNA into ssDNA.**
4. A solution of **radioactively-labelled gene probe** is added so that it **will hybridize with the complementary DNA sequence** of the **human growth hormone gene**.
5. **Wash** away **excess probe**
6. The position of the cell containing the DNA-gene probe hybrid/labelled gene can be located by **autoradiography** performed on the filter.
7. The position of **dark spots** on the x-ray film corresponds to the **position of the bacterial colonies containing the human growth hormone gene**.

   Accept: Fluorescent-labelled gene probes and subsequent corresponding steps.

(e) Suggest an explanation for the inclusion of the polyA signal sequence in pCMV6-XL5. [2]

- If the plasmid is inserted into a **eukaryotic host cell**, the polyA signal sequence will **direct the addition of a poly A tail to mRNA**.
- Polyadenylation of mRNA **extends the half-life of mRNA** / **increase mRNA stability** / reduces the rate at which mRNA is hydrolysed by endonucleases, resulting in more proteins synthesis / translation can occur for a longer period of time.

  Facilitate the export of mRNA out of the nucleus into the cytoplasm.

(f) Plasmid replication is usually initiated by RNAII, which is a sequence of ribonucleotides that binds to plasmid DNA prior to replication.

State and explain the role of RNAII in plasmid replication. [2]

- RNAII serves as a **primer** for DNA replication.
- Because **DNA polymerase** requires a **3'OH** to bind to in order to start/begin replication.

[Total: 15]

QUESTION 2

The location of the gene locus responsible for cystic fibrosis was not discovered until 1985. Scientists used restriction fragment length polymorphism to discover the genetic markers associated with the disease. One such marker was a 950bp-long region known as KM19 that had been sequenced prior to 1985.

Samples of DNA were obtained from a family known to have the condition. The KM19 locus was amplified by PCR and mixed with *Pst*I and *Eco*RI in two separate restriction digests. The results of gel electrophoresis of both restriction digests are shown in Fig. 2.1.

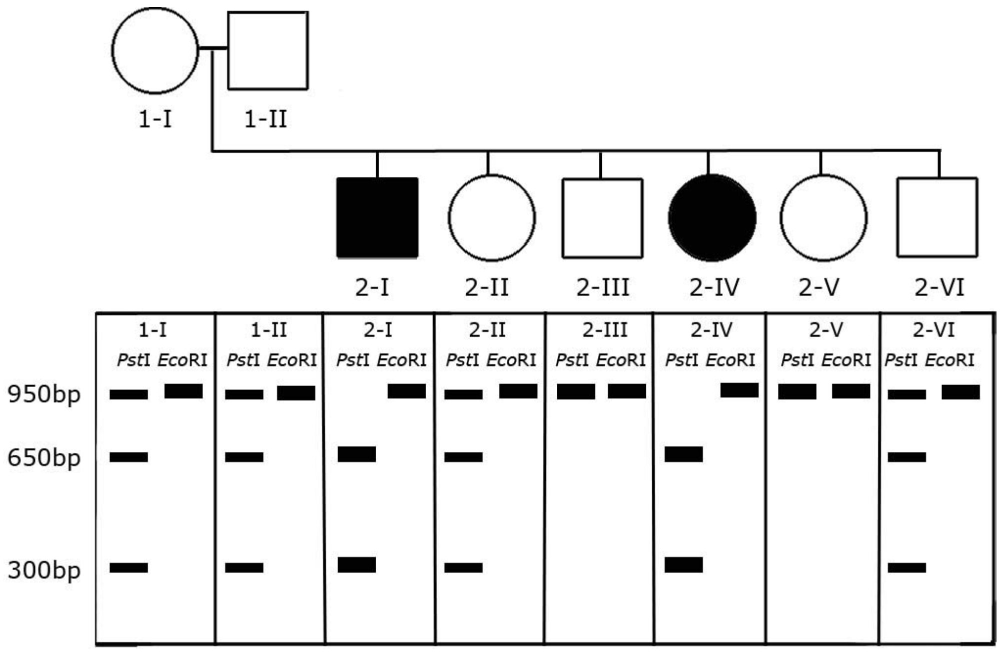


Fig. 2.1

(a) Using the information in Fig. 2.1,

(i) state and explain which restriction enzyme digest should be used to detect the KM19 genetic marker associated with cystic fibrosis. [3]

- ***Pst*I**
- Digestion with *Pst*I produced **two bands** of **300bp and 650bp** in **affected individuals** but **unaffected individuals** showed either a **single band of 950bp** or **three bands of 950bp, 650bp and 300bp**.
- But after digestion with *Eco*RI, **all individuals** had **single 950bp fragments** and the affected and unaffected individuals are not differentiated.

(ii) draw a restriction map of the KM19 genetic marker that is associated with cystic fibrosis. [1]

- [Size of fragment + labelling of restriction site]

(b) Explain why genetic markers like KM19 can be used to detect the presence of disease-causing alleles. [2]

- These genetic markers are **polymorphic**
- They are **tightly linked** to the disease-causing alleles on the same chromosome and are likely to be **inherited together** as one unit with the disease-causing allele.

(c) Further research in the 1980s identified sequences of genetic markers located both upstream and downstream of the gene suspected to cause cystic fibrosis.

Using this information, suggest how the gene could be isolated using these two genetic markers. [2]

- Design **primers** that are **complementary** to the **genetic markers** upstream and downstream of the suspected gene.
- Use the **polymerase chain reaction** (PCR) to **synthesise multiple copies** of suspected gene.

Clinical trials for the treatment of inherited conditions via gene therapy have been conducted in the past two decades, but an unsuccessful and lethal trial in 1999 revealed the risks involved in using adenoviruses in gene therapy. Adeno-associated viruses were then used as alternative viral vectors for gene therapy trials as they exhibit little or no disease-causing side effects. More recently, non-viral gene therapies have been explored as safer alternatives to viral vectors.

(d) Describe two other advantages of utilising adeno-associated viruses as compared to adenoviruses. [2]

- Adeno-associated viruses are capable of infecting both **non-dividing and dividing cells**.
- Adeno-associated viruses are capable of integrating the normal allele into the patient's genome at a **specific site**, hence there is a **lesser chance of oncogene formation**.
- Treatment is **permanent** if **stem cells** are used as the normal allele is integrated into the cell's genome and the stem cells are capable of self-renewal.

(e) State two criteria that make cystic fibrosis a condition that can be potentially treated with gene therapy. [2]

- CF is a **homozygous recessive condition** hence insertion/introduction of the normal dominant allele that can produce / code for a functional protein.
- CF is a **single-gene disorder** hence it can be treated by the introduction of the normal allele by gene therapy.

Immune responses to non-viral gene therapy have been documented. The main cause of the immune response was found to be the plasmids carrying the normal allele.

(f) Suggest a reason why plasmids may elicit an immune response during non-viral gene therapy. [1]

- Plasmids used for gene therapy may have **sequences** that are **bacterial/prokaryotic in origin** and are **recognised by the human immune system**.

(g) State three other differences between non-viral gene therapy and viral gene therapy. [3]

Feature	Non-viral	Viral
Vector(s) used	Examples of non-viral vectors: liposomes, cationic polymers	Adenoviral vectors, retroviral vectors, adeno-associated viral vectors.
Potential mutation	Less risk of mutations as most methods do not involve integrating the normal allele into the genome of the patient's cells.	Greater risk of harmful mutations as a result of viruses inserting genes randomly into the patient's genome.
Size of insert	Larger size of gene can be inserted.	Only a limited length of gene can be inserted.
AVP	AVP	AVP

[Total: 16]

QUESTION 3

Embryonic stem (ES) and induced pluripotent stem (iPS) cells hold great promise for the treatment of a wide variety of acquired or hereditary diseases. One of the major obstacles to clinical applications is directing them to differentiate into functional cells.

The effect of various growth factors on stem cell differentiation in ectodermic, mesodermic and endodermic embryonic stem cells was carried out and the process was documented in Fig. 3.1. Embryonic stem cells were cultured and embryoid bodies allowed to differentiate into ectodermic, mesodermic and endodermic embryonic stem cells. Growth factors were added to the stem cells and the expression of cell-specific genes was measured after ten days.

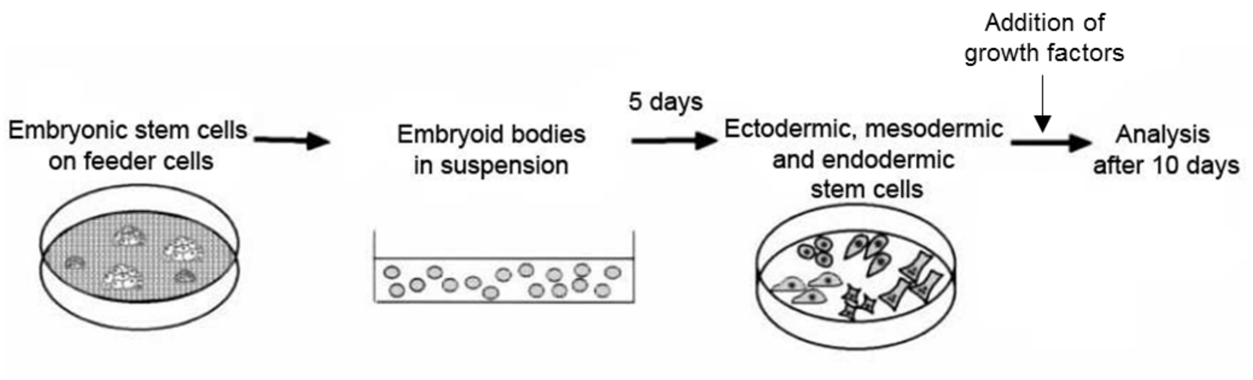


Fig. 3.1

One of the growth factors tested was the epidermal growth factor. 100ng/mL of epidermal growth factor was given to the three different embryonic stem cells. The results are summarised in Table 3.1.

Type of stem cell	Expression of cell-specific genes after ten days		
	Keratin	α-Globin	Insulin
Embryonic Ectoderm	+	–	–
Embryonic Mesoderm	–	+	–
Embryonic Endoderm	–	–	–

\+ genes are expressed
– genes are not expressed

Table 3.1

(a) Using the information given and the data from Table 3.1, describe and explain the effect of epidermal growth factor on the expression of keratin in stem cells. [3]

- Addition of 100ng/mL of epidermal growth factor results in <u>the **expression of keratin in ectodermic stem cells**</u>, but <u>**not mesodermic or endodermic stem cells**</u> after ten days.
- <u>**Epidermal growth factor receptor**</u> was <u>**present only on ectodermal stem cells**</u>.
- Epidermal growth factor <u>**initiates the cell-signalling pathway**</u>…
- …leading to the <u>**transcription and translation**</u> of the <u>**keratin gene**</u>.

(b) Suggest why the insulin gene was not expressed in the three different kinds of embryonic stem cells even after treatment with a variety of other growth factors. [1]

- The insulin gene will only be transcriptionally active in β-cells / will not be synthesised until the pancreas is fully developed.

Researchers have recently used induced pluripotent stem (iPS) cells generated from human skin cells to create precursor heart cells.

(c) State two advantages of using iPS cells instead of embryonic stem cells for research and clinical trials. [2]

- iPS cells can be generated from skin cells/adult cells, so there are **fewer ethical issues** involved as compared to using embryonic stem cells obtained from an embryo.

- iPS cells can be generated from skin cells/adult cells of the patient/sufferer, so there might be **less risk of tissue rejection** after gene therapy and transplantation.

Cord blood stem cell therapy has been used in clinical trials for the treatment of autoimmune diseases like leukaemia and inherited conditions like Severe Combined Immuno-deficiency (SCID).

The two main modes of treatment are:

- o Allogeneic – The patient receives stem cells from a matching donor, either a sibling or an unrelated donor

- o Autologous – The patient receives their own stem cells

(d) Describe the properties of cord blood stem cells. [2]

- Multi-potent and able to undergo differentiation to produce different types of blood cells.
- Unspecialised and undifferentiated.
- Has telomerase activity.
- Has ability to undergo long-term self-renewal.

(e) Suggest a potential limitation of autologous cord blood stem cell therapy. [1]

- Cord blood stem cell therapy cannot cure inherited genetic diseases if the donor has the same inherited condition or the treatment is autologous (as the patient receives his own stem cells).

- The amount of cord blood stem cells that can be collected from a single donor is small and reduces chances of repeated therapy.

[Total: 9]

Question 4 Planning Question

You are required to plan, but not carry out, an investigation into the effect of temperature from 10°C to 70°C on the permeability of potato cell membranes. Potatoes are rich in potassium ions. When small disc-shaped potatoes cut from a core borer (Fig. 4.1) are placed in water, potassium ions are released from the cell into the water.

The concentration of potassium ions in the water can be measured using an electrode (Fig. 4.2) that is selective for potassium ions.

Fig. 4.1

Fig. 4.2

Your plan should:

- Have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it.
- Be illustrated by relevant diagram(s) to show, for example, the arrangement of apparatus used.
- Include the layout of result tables and graphs with clear headings and labels.
- Include full details and explanations of the procedure that you would adopt to ensure that the results obtained were as quantitative, precise and reliable as possible.

Your planning must be based on the assumption that you have been provided with the following equipment and apparatus which you must use:

- Potatoes
- Potassium ion-selective electrode which measures in mg/L
- A core borer of 10mm in diameter
- Scalpel
- Ruler
- Boiling tubes
- Measuring cylinders
- Stopwatch
- Thermostatically-controlled water bath
- Thermometer
- Isotonic buffer solution
- De-ionized water
- Other common laboratory apparatus

[Total: 12]

Objective
To investigate the effect of temperature on the permeability of cell membranes using potatoes

Background and hypothesis
- Ions are charged
- Ion channels to provide hydrophilic channel
- Lateral movement of phospholipids
- Fluidity of membrane
- Increase temperature, more heat energy, phospholipids gain kinetic energy
- Membrane more fluid and leaky
- At high temperatures, membrane disrupted, K^+ fully released into water
- Cell wall is fully permeable to ions
- Higher temperature, greater membrane permeability, more K^+ diffuse out

Theory on membrane [1]

Independent and dependent variables
Independent variable
- Temperature / °C
- 10, 20, 30, 40, 50, 60, 70°C

Dependent variable
- Concentration of K^+ / mgL^{-1}

Correct variables with units [1]
At least 5 temperatures of sensible range and interval [1]

Other variables to be kept constant
- Same size and number of potato discs from the same potato
- Volume of water in all tubes
- Fresh de-ionized water in all tubes
- Time for testing
- Same electrode

Any two with an indication of how it is kept constant in the methods [1]

Control
- For every temperature examined, no potato disc added – to ensure the K+ detected is from the potato cells.

Description of control with rationale [1]

Methods
1. Cut a potato into cylindrical shapes of diameter 10mm using the core borer.
2. Cut the cylindrical potato into disc shape of height 3mm using scalpels.
3. Place the disc-shaped potatoes into an isotonic buffer. This is to prevent the potato from drying up and also to maintain its osmotic pressure.
4. Add 10ml of fresh de-ionized water into a boiling tube and equilibrate in a 10°C water bath until the de-ionized water is at 10°C. Monitor temperature using thermometer.
5. Add 10 discs of potatoes and start the stopwatch.
6. After 15min (accept range), measure the concentration of K^+ with the electrode.
7. Repeat step 4 – 6 twice to obtain a total of 3 readings to minimize error by calculating average.
8. Repeat step 4 – 7 for 20, 30, 40, 50, 60 and 70°C.
9. Repeat the whole experiment twice to ensure reproducibility.

Obtaining potato discs (point 1 – 3) [1]
Measurement of readings (point 4 – 6, 8) [1]
Performing replicates and repeats (point 7, 9) [1]

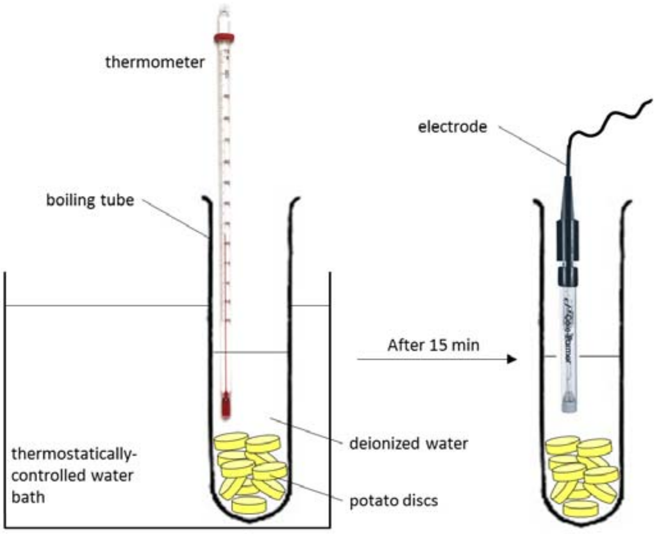
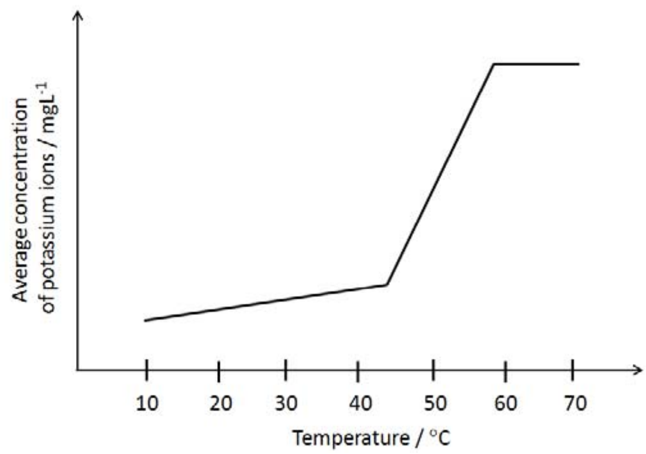


Diagram of apparatus set-up [1]

Results

Temperature / °C	Concentration of K^+ / mgL^{-1}			
	1	2	3	Average
10				
20				
30				
40				
50				
60				
70				

Table with appropriate headings with units [1]



Labeled axes and correct proposed trend [1]

Risks and precautions

Risk	Precaution
Scalpel is sharp and may cause injuries	Take extra care when handling with the scalpel
70°C water bath is hot enough cause scalding	Use cloth/ for insulation

Risk with corresponding precaution [1]

QUESTION 5

(a) Explain how genetically modified organisms may provide a solution to global food shortage issues. [8]

1. **Increase yield** by genetically modifying food crops that will result in decreased losses caused by pesticides, fungal infestation, viruses, bacterial infection and pests.

Introduction of genes that code for proteins that confer insecticidal resistance

2. Example: Genetically engineered crop plants that express the **Bt-toxin gene** from the bacteria *Bacillus thuringiensis* produce **Bt-toxin protein** which **kills insect larvae feeding on the plant**.

Introduction of genes that code for proteins that confer herbicidal resistance

3. Example: Genetically engineered crop plants that are **resistant to herbicides** by introducing the **herbicide/glyphosate resistance gene**

4. When herbicide is applied to the field, the **weeds are eliminated** **but not the herbicide resistance crop plants**.

Introduction of genes that code for proteins that confer viral resistance

5. E.g., tobacco plant can be made **resistant to the tobacco mosaic virus** by expressing the **coat protein gene** of a virus.

Introduction of genes that code for proteins that confer fungal resistance

6. Example: Genetically engineered crop plants have an **increased resistance to fungal infestations** due to the expression of **anti-fungal proteins** within the plant.

   Or

   Inclusion of a gene expressing anti-fungal compounds (e.g. phytoalexin) into transgenic crop plant (e.g. Alfafa, potato) confer fungal resistance.

Introduction of genes that code for growth hormones

7. Example: The Atlantic salmon has been genetically modified by the addition of a **growth hormone gene** from a Pacific Chinook salmon and an active promoter from an ocean pout placed upstream of the growth hormone gene.

8. The insertion of the growth hormone gene results in **faster growth rate** and yield of the salmon, thereby increasing the supply of salmon.

Introduction of genes that delayed ripening

9. Flavr Savr tomatoes are engineered to include a gene for **antisense mRNA** to **polygalacturonase gene** to **reduce expression of polygalacturonase**.

10. This will result in **delayed ripening** allowing **crops to be stored for longer period** of time

(b) Discuss the social and ethical implications of germ-line gene therapy. [7]

Limitation in technology

1. The technology for germ-line gene therapy is not perfect and there is a risk of inducing mutations, which can be passed down to the offspring, when the insertion of gene happens to introduce further defects.

Research on embryos

2. Those involved in the research related to germ-line gene therapy regularly create and destroy embryos as a part of their research.
3. There are objections to killing embryos used for research in gene therapy as human life may be considered to have begun at conception.

Abuse of technology

4. It may be abused by parents who want to eliminate undesirable alleles from their offspring.

Denial of human rights

5. The individuals that results from germ-line gene therapy did not have a say in whether their genetic material should have been changed.

The fate of human descendants

6. The genetic makeup of human descendants may be changed hence affecting the human gene pool.
7. These changes may presumably be for the better. However, any errors in technology and judgement may have far-reaching consequences.

Ethical and social objections of therapy in general

8. ***The definition of 'normal' and 'disability' may change.*** The decision of what is normal and what is a disability should not rest on just a small group of individuals or medical professionals. The question of whether disabilities are considered disease and whether these need to be cured or prevented requires thorough consideration.

9. ***Demeaning the life of the disabled.*** There is a concern that the presence of a cure might demean the lives of individuals presently affected by disabilities. There could be a new definition of what contributes to a 'normal' being. E.g. the exclusion of individuals who are merely average in intelligence.

10. ***Creation of social class distinctions.*** There can be a formation of class distinctions, because expensive gene therapy is easily available to individuals with financial means, but not to those financially less well off. There should also be an establishment of procedural fairness in selection of patients for research.

11. **Religious and cultural belief.** Changing the genetic makeup of individuals goes against many religious and cultural beliefs.

12. ***Other issues.*** There may be incidental elimination of genes that are important but with unknown functions and this may lead to decreased species fitness

(c) Discuss the goals of the Human Genome Project. [5]

1. Identify all the genes in human DNA (approximately 20000 -25000).

2. Determine the DNA sequence of the entire human genome (about 3 billion base pairs).

3. Store this information in databases to be shared publicly by researchers throughout the world

4. Improve tools for data analysis.

5. Transfer related technologies to the private sector and award grants for innovative research, so as to catalyse the biotechnology industry and the development of new medical applications.

6. Address the ethical, legal and social issues that may arise from the project.

• End of Paper •